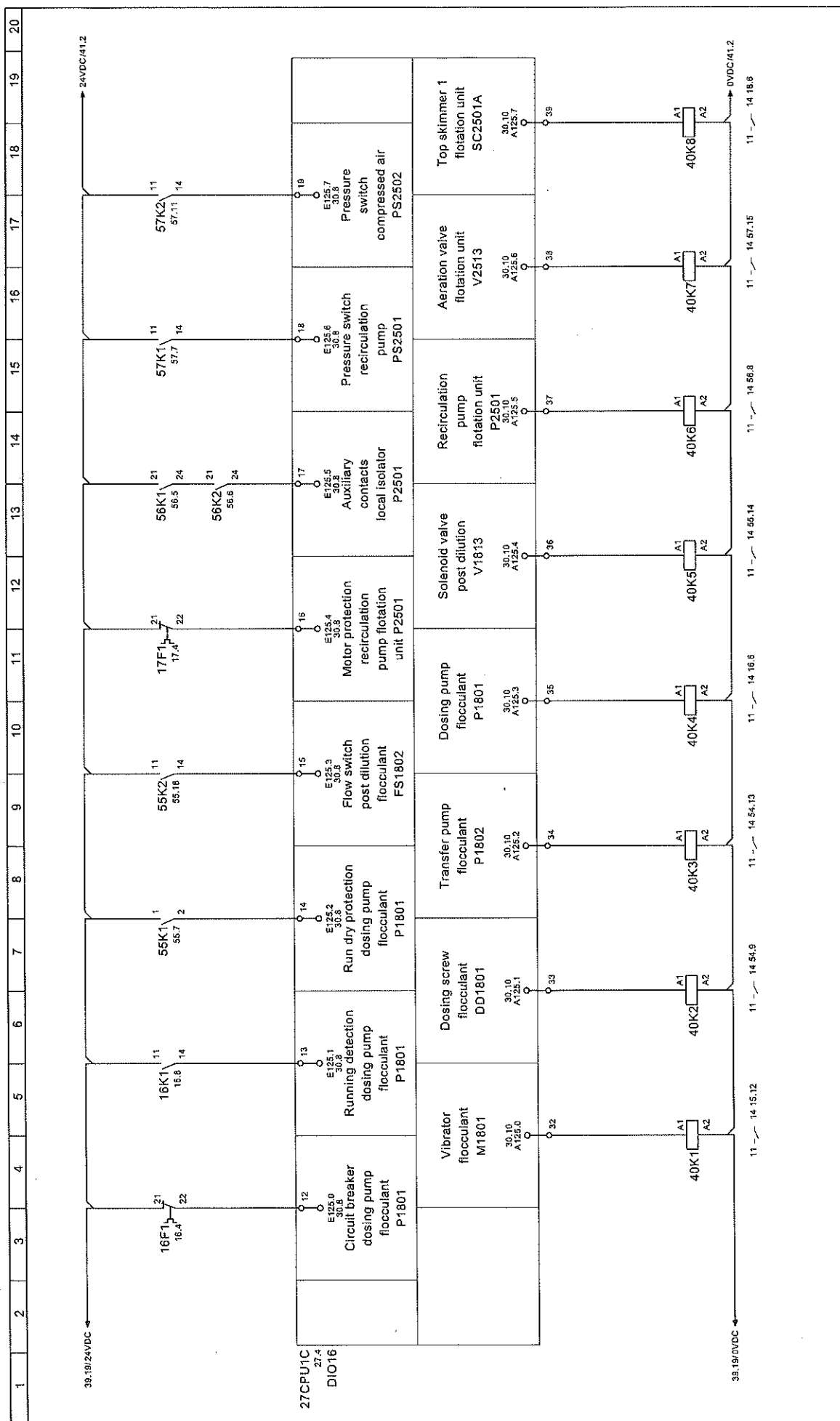
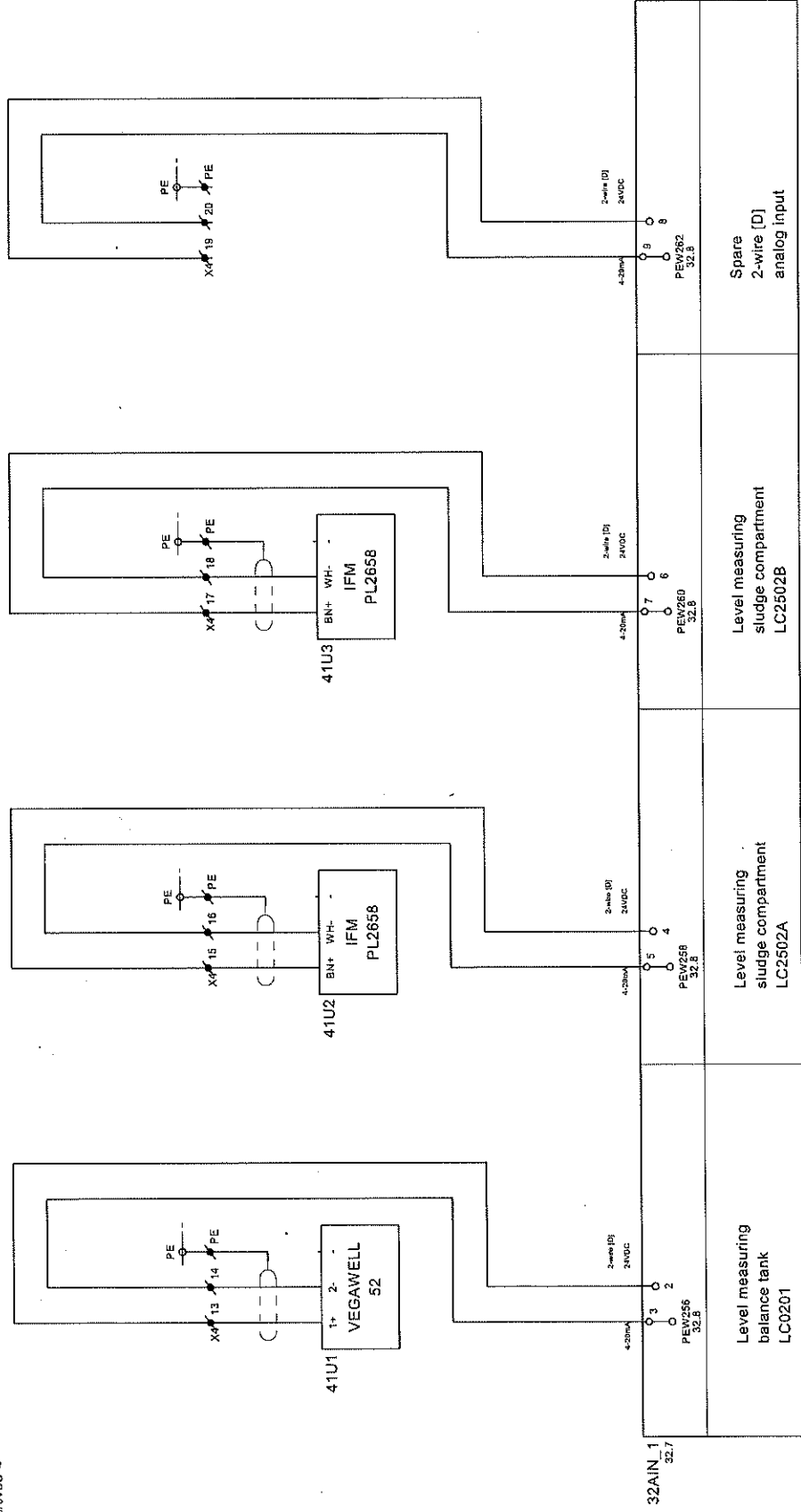




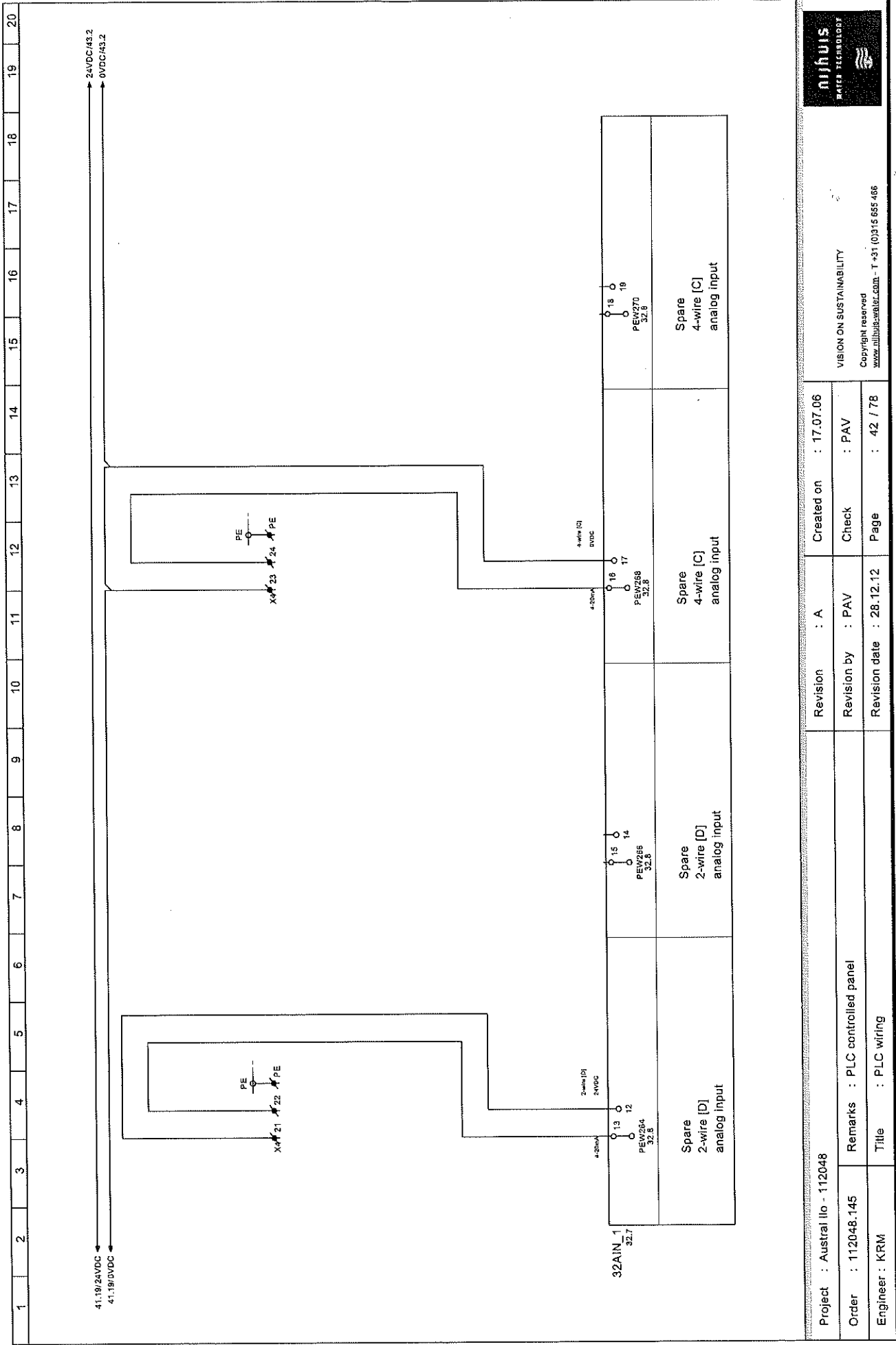
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40.19/24VDC ← 24VDC/42.2
 40.19/0VDC ← 0VDC/42.2



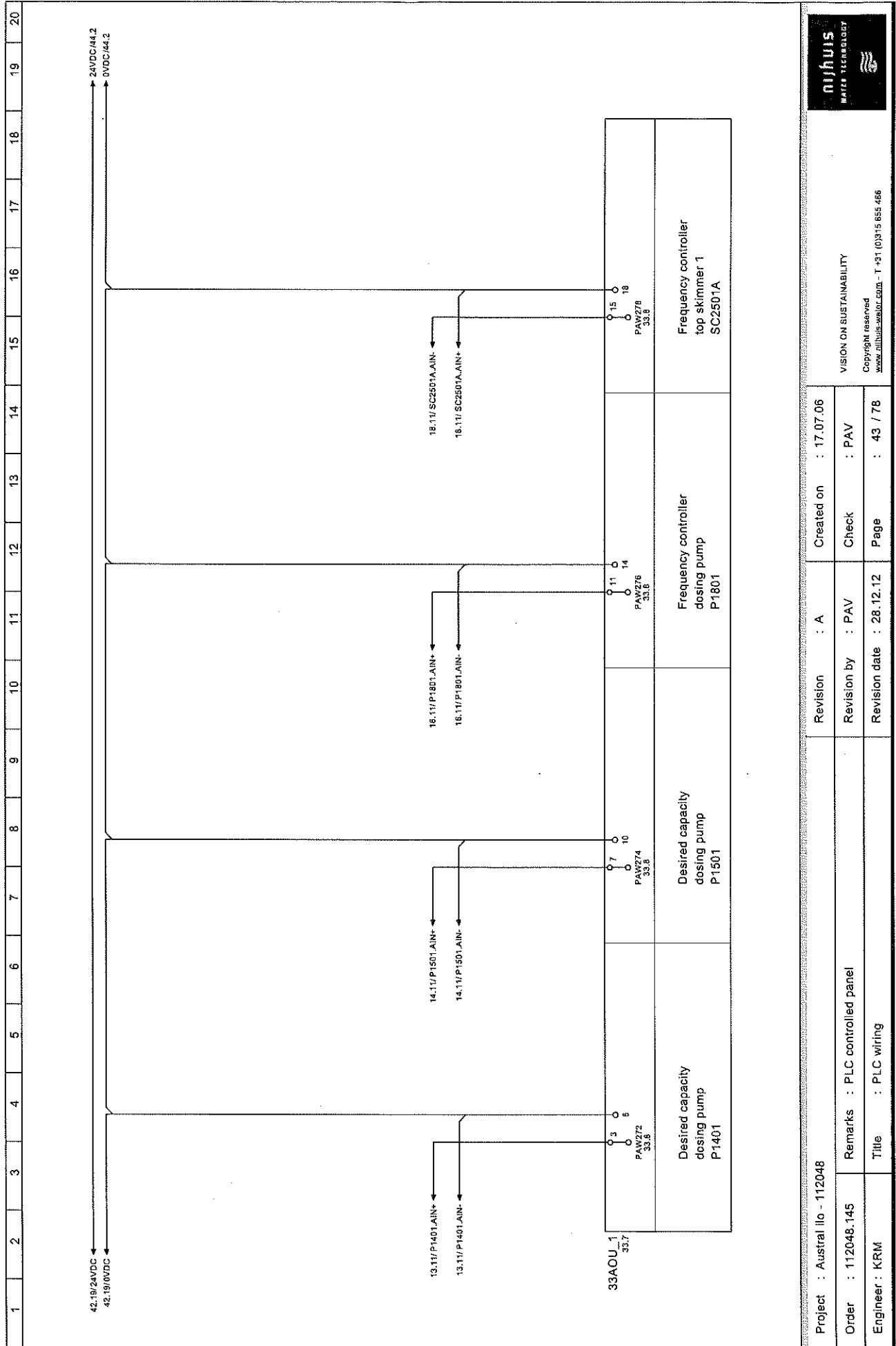
Project : Austral Ilo - 112048		Revision : A	Created on : 17.07.06
Order : 112048.145		Revision by : PAV	Check : PAV
Engineer : KRM		Revision date : 28.12.12	Page : 41 / 78
Remarks : PLC controlled panel		Title : PLC wiring	





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Project : Austral Ilo - 112048	Revision : A	Created on : 17.07.06
Order : 112048.145	Revision by : PAV	Check : PAV
Engineer : KRM	Revision date : 28.12.12	Page : 42 / 78
Remarks : PLC controlled panel	Title : PLC wiring	



Project : Austral Ilo - 112048

Order : 112048.145

Remarks : PLC controlled panel

Engineer : KRM

Title : PLC wiring

Revision : A

Revision by : PAV

Revision date : 28.12.12

Created on : 17.07.06

Check : PAV

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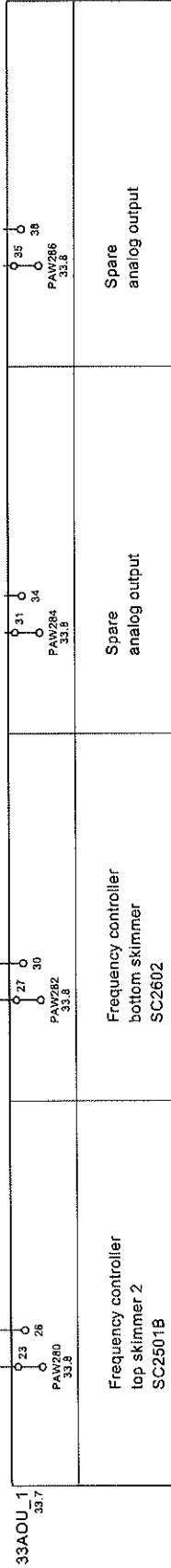
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43.18/24VDC → 24VDC/45.2
43.19/0VDC → 0VDC/45.2

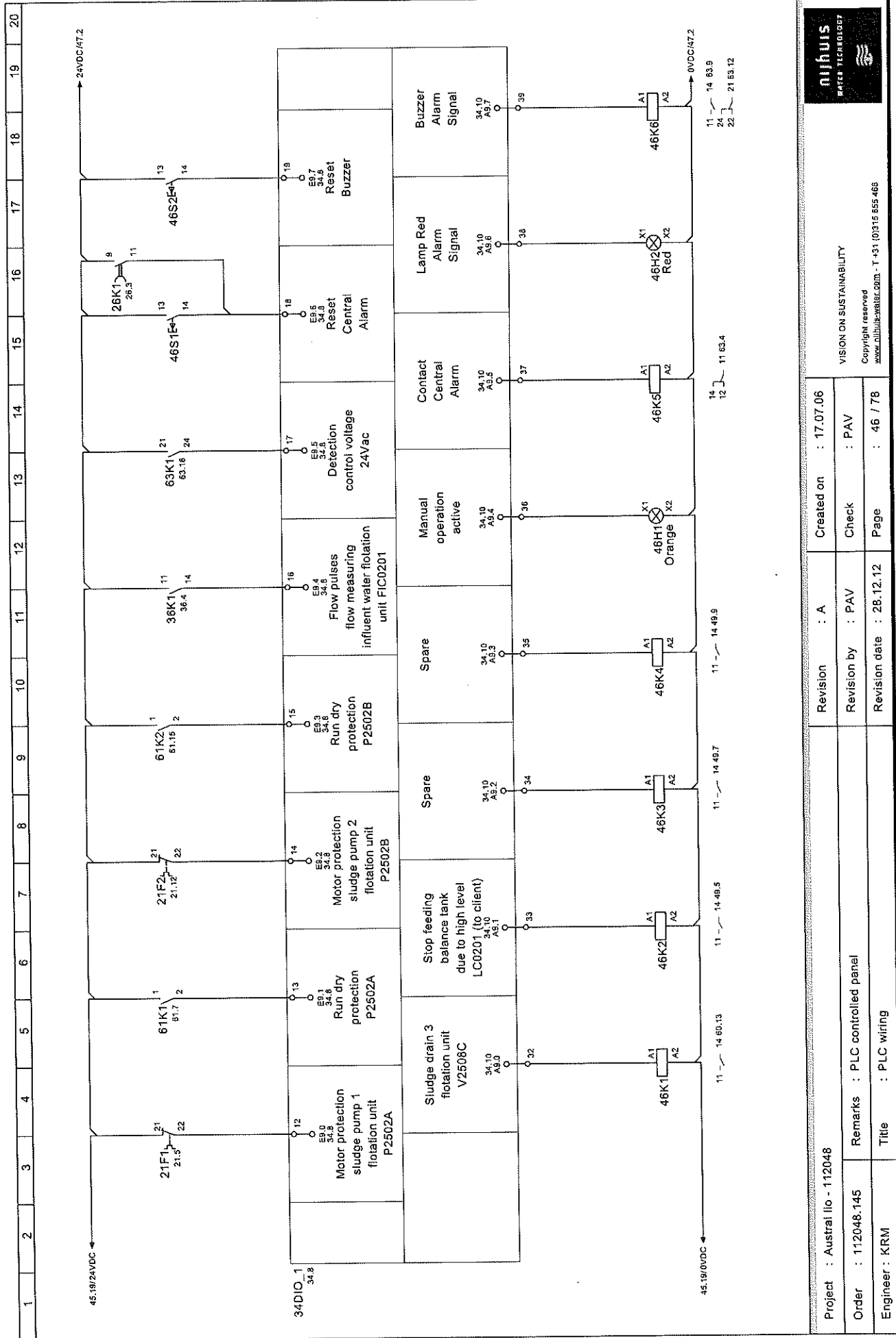
18.11/ SC2501B_AIN- → 20.11/ SC2502_AIN+
19.11/ SC2501B_AIN+ → 20.11/ SC2502_AIN+



Project : Austral Ilo - 112048		Revision : A	Created on : 17.07.06
Order : 112048.145	Remarks : PLC controlled panel	Revision by : PAV	Check : PAV
Engineer : KRM	Title : PLC wiring	Revision date : 28.12.12	Page : 44 / 78

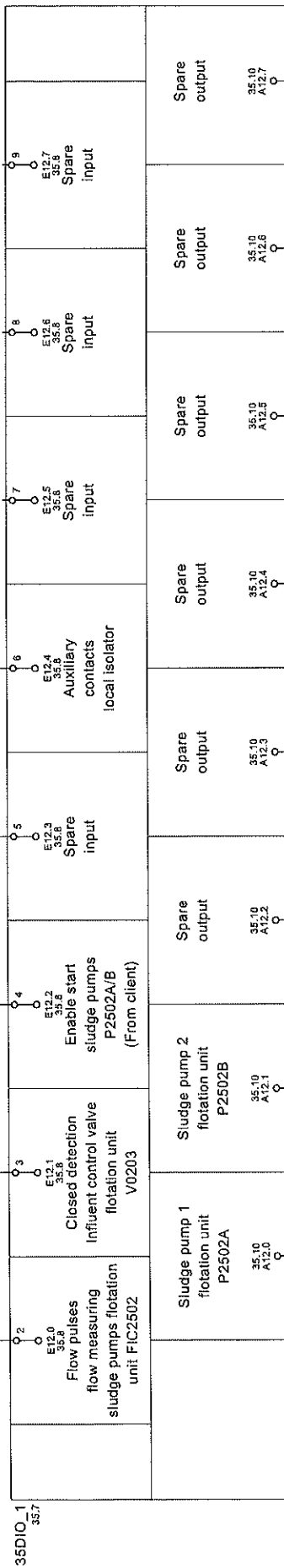


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45.19/24VDC ← 24VDC/48.2 →



45.19/0VDC ← 0VDC/48.2 →

11 14 61.5 11 14 61.5

NEW

Project : Austral Ilo - 112048

Order : 112048.145 Remarks : PLC controlled panel

Engineer : KRM Title : PLC wiring

Revision : A

Revision by : PAV

Revision date : 13.03.13

Created on : 17.07.06

Check : PAV

Page : 47 / 78

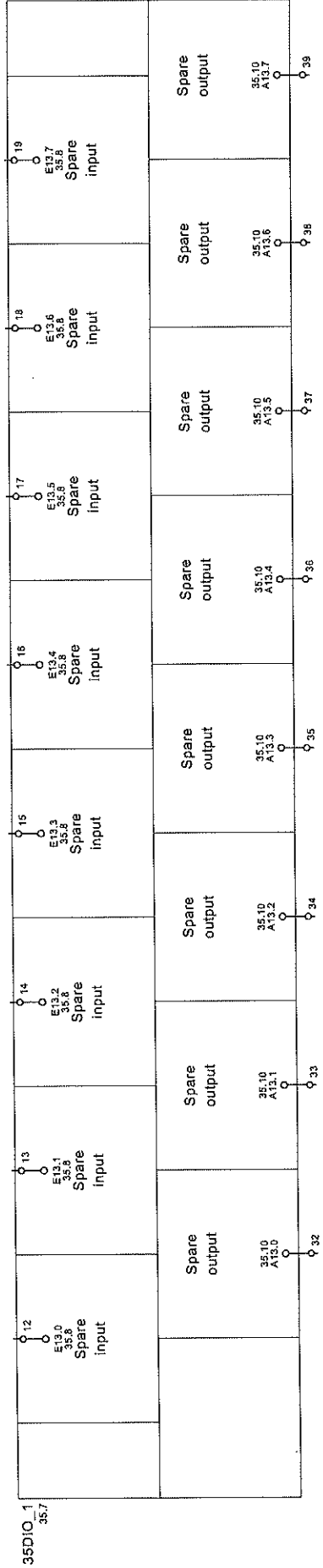
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47.19/24VDC ← 24VDC/57.3

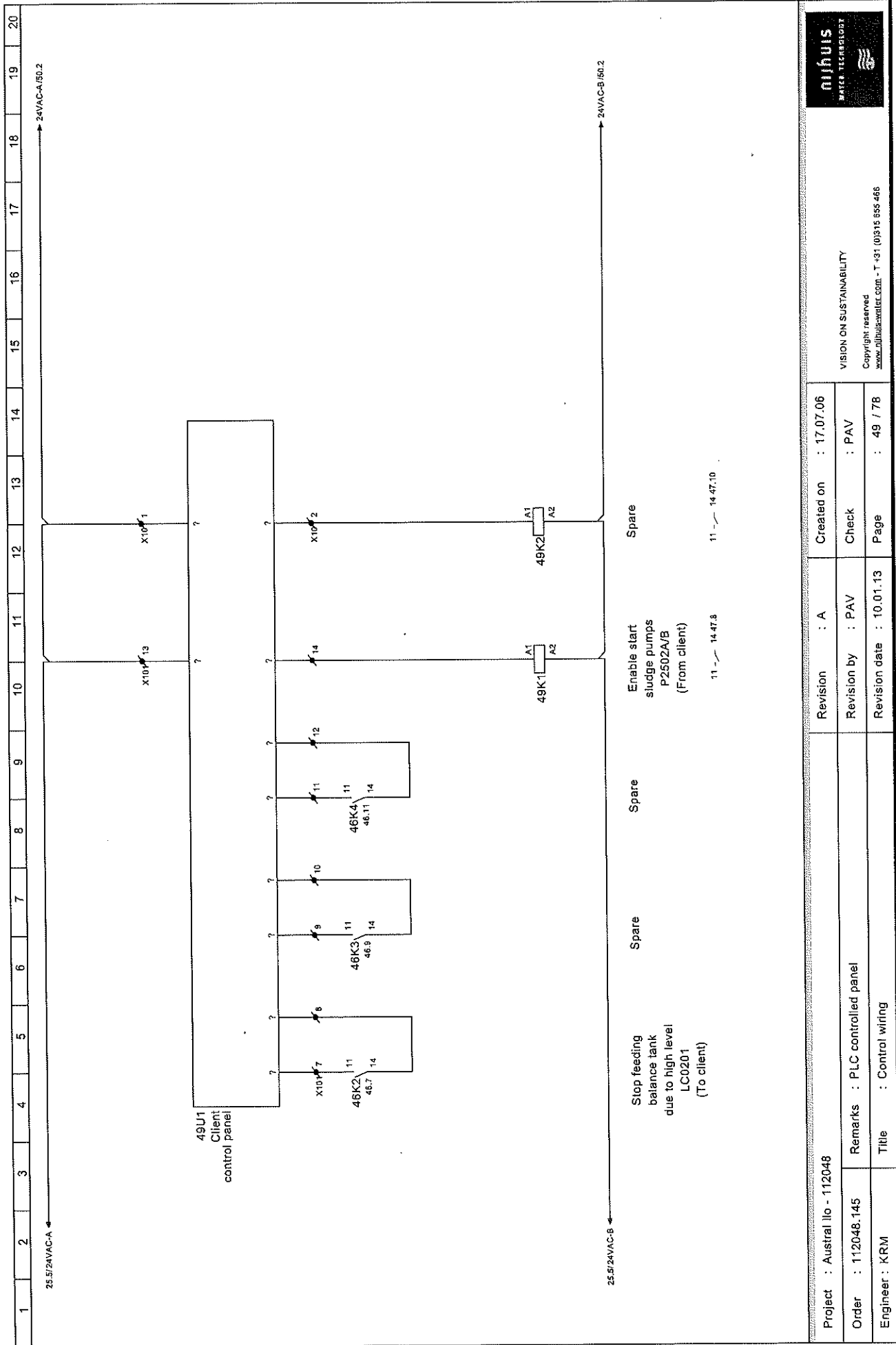


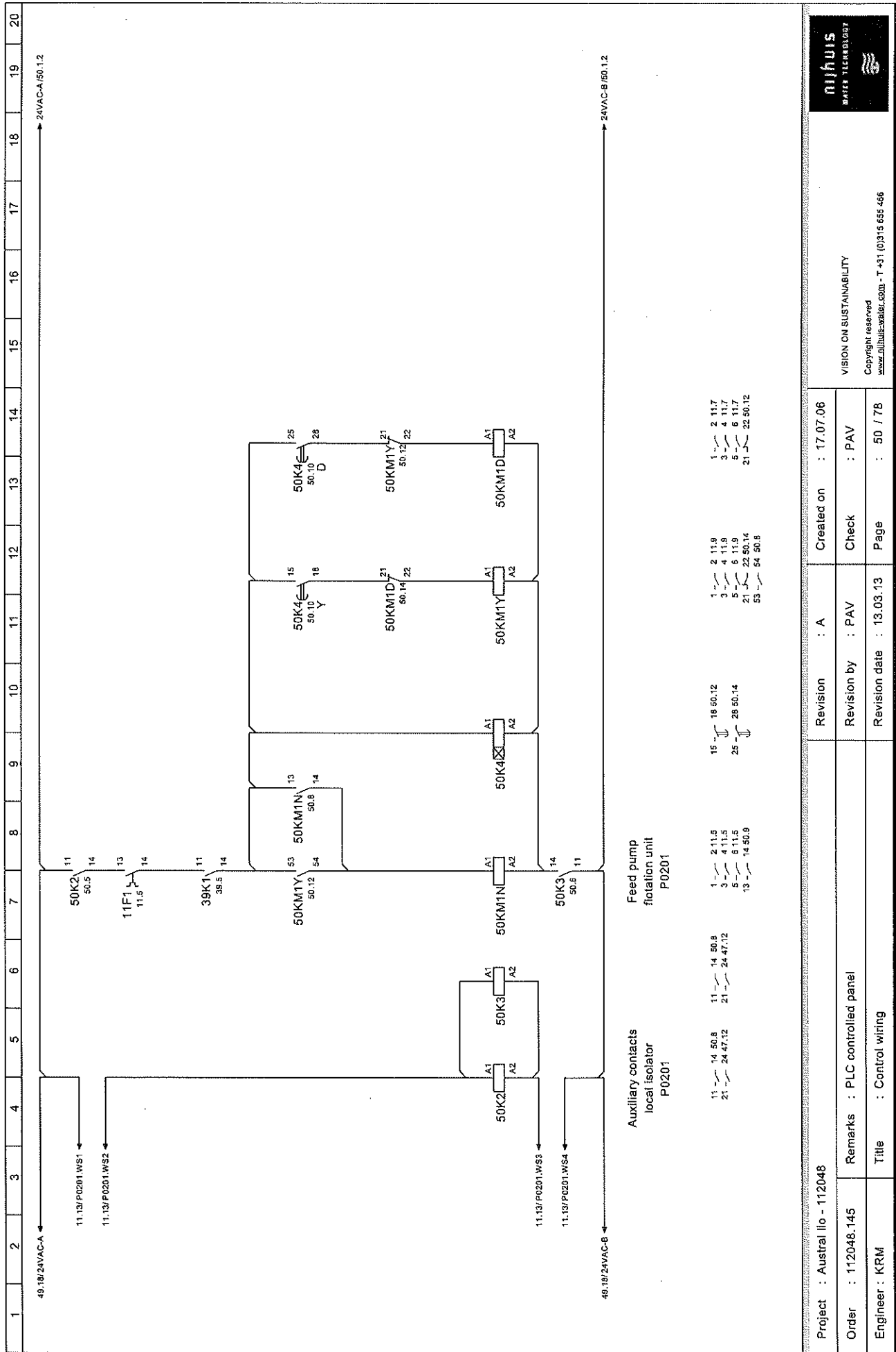
47.19/0VDC → 0VDC/57.3

Project : Austral Ilo - 112048		Revision : A	Created on : 17.07.06
Order : 112048.145	Remarks : PLC controlled panel	Revision by : KRM	Check : PAV
Engineer : KRM	Title : PLC wiring	Revision date : 06.04.12	Page : 48 / 78



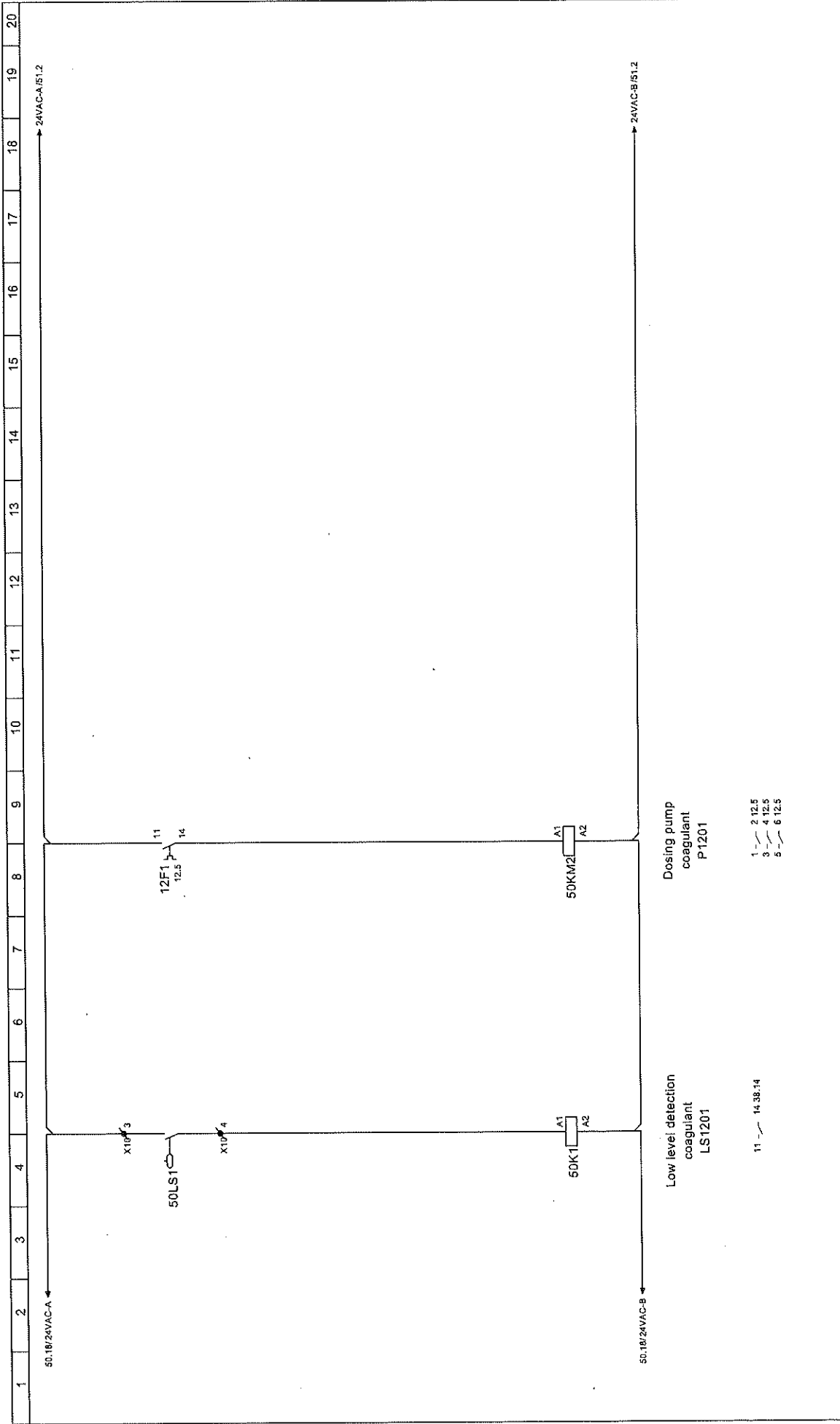
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Project : Austral Ilo - 112048	Revision : A	Created on : 17.07.06
Order : 112048.145	Revision by : PAV	Check : PAV
Engineer : KRM	Revision date : 13.03.13	Page : 50 / 78
Remarks : PLC controlled panel		
Title : Control wiring		

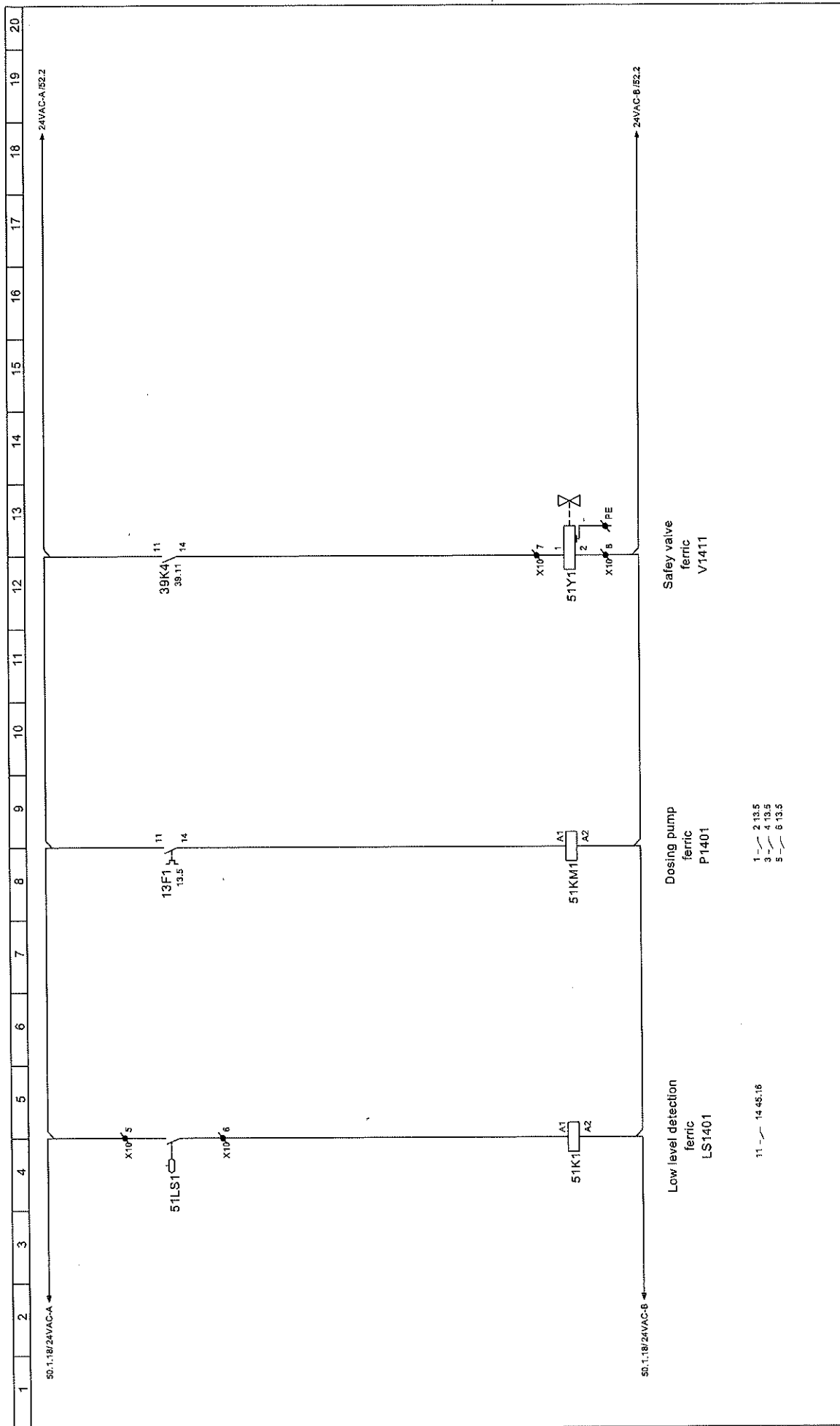


Project : Austral Ilo - 112048		Revision : A	Created on : 17.07.06
Order : 112048.145	Remarks : PLC controlled panel	Revision by : RKH	Check : PAV
Engineer : RKH	Title : Control wiring	Revision date : 10.01.13	Page : 50.1 / 78

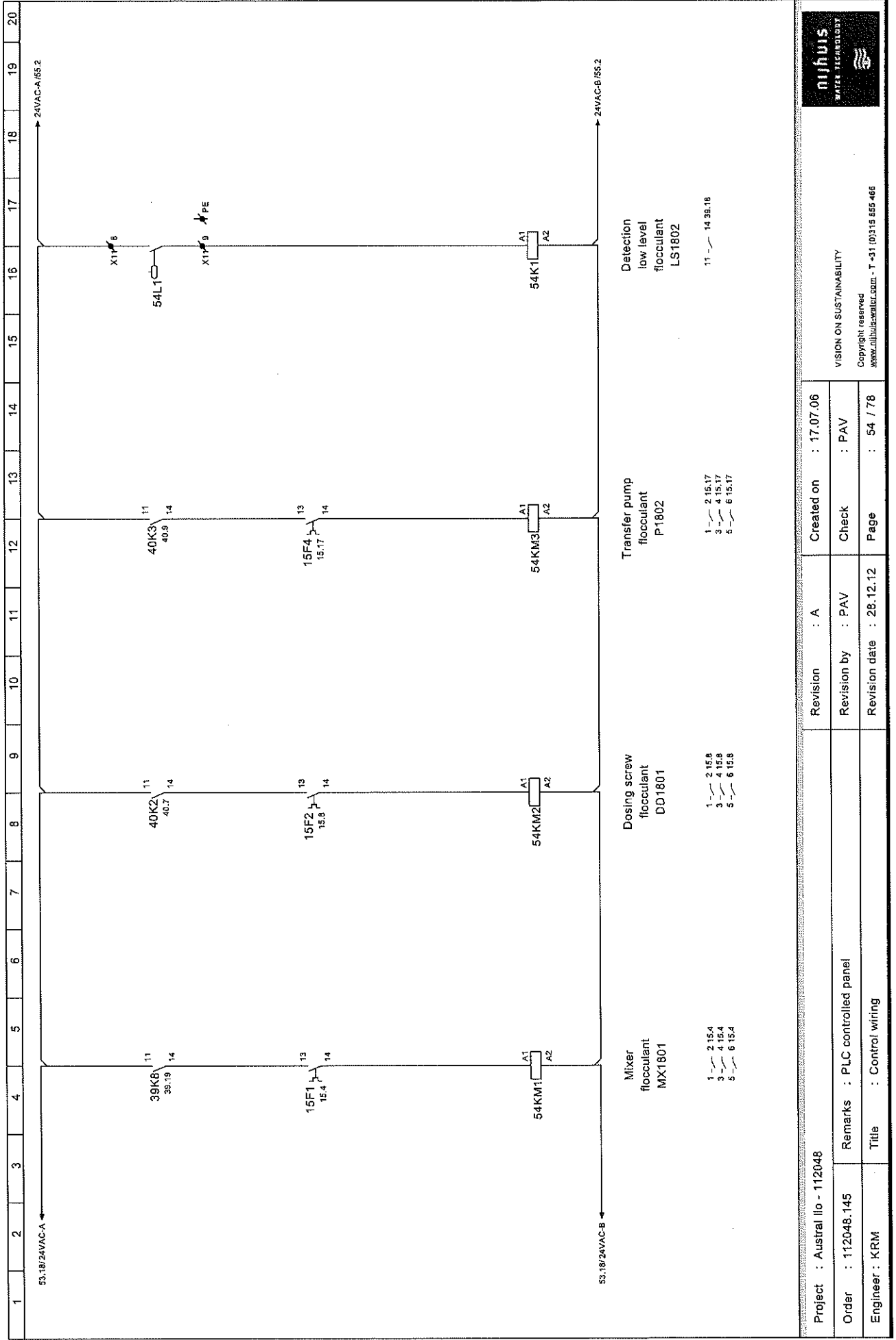


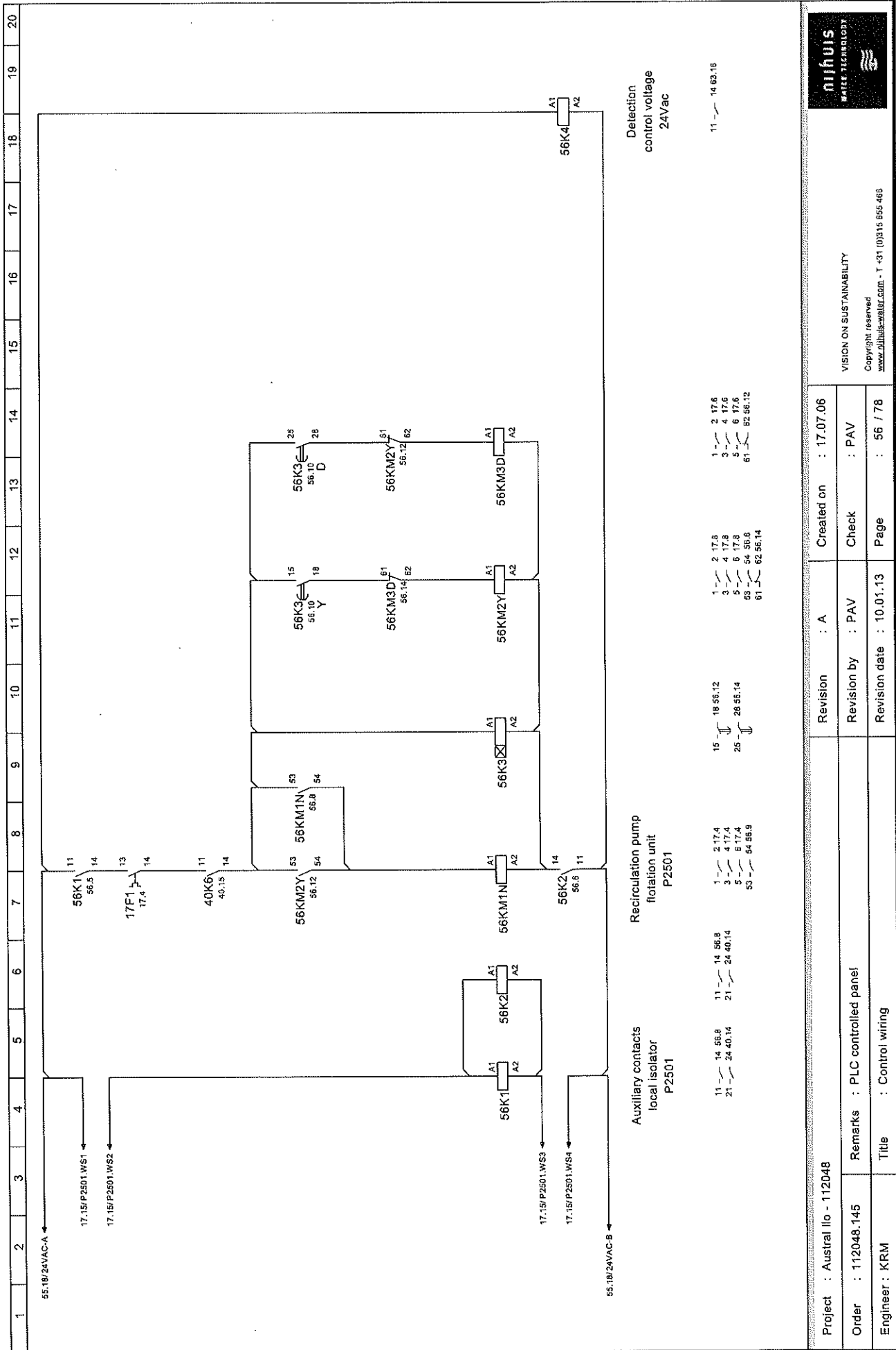
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Project : Austral Ilo - 112048		Revision : A	Created on : 17.07.06
Order : 112048.145	Remarks : PLC controlled panel	Revision by : RKH	Check : PAV
Engineer : KRM	Title : Control wiring	Revision date : 10.01.13	Page : 51 / 78

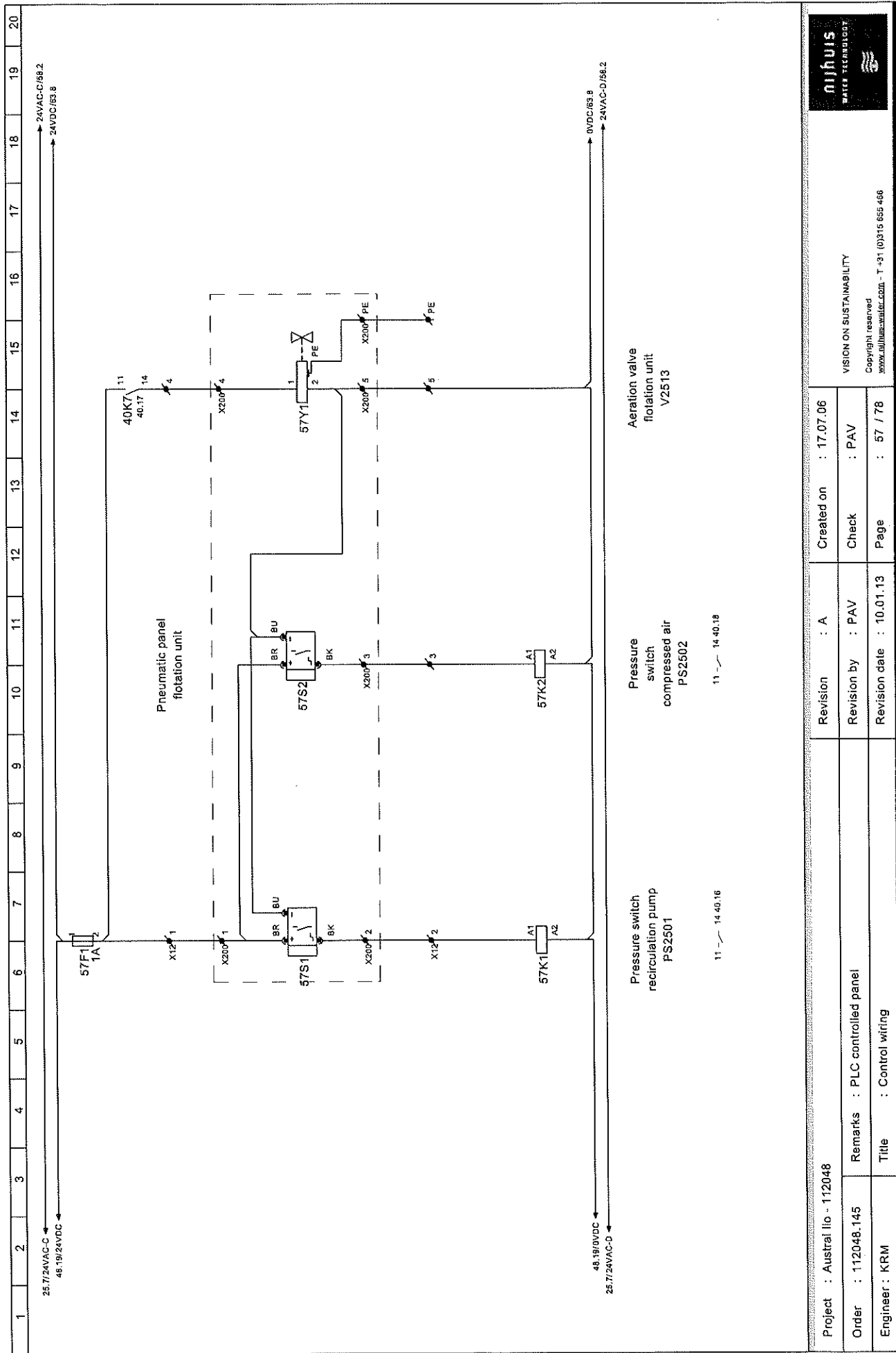




Project : Austral Ilo - 112048		Revision : A	Created on : 17.07.06
Order : 112048.145	Remarks : PLC controlled panel	Revision by : PAV	Check : PAV
Engineer : KRM	Title : Control wiring	Revision date : 10.01.13	Page : 56 / 78



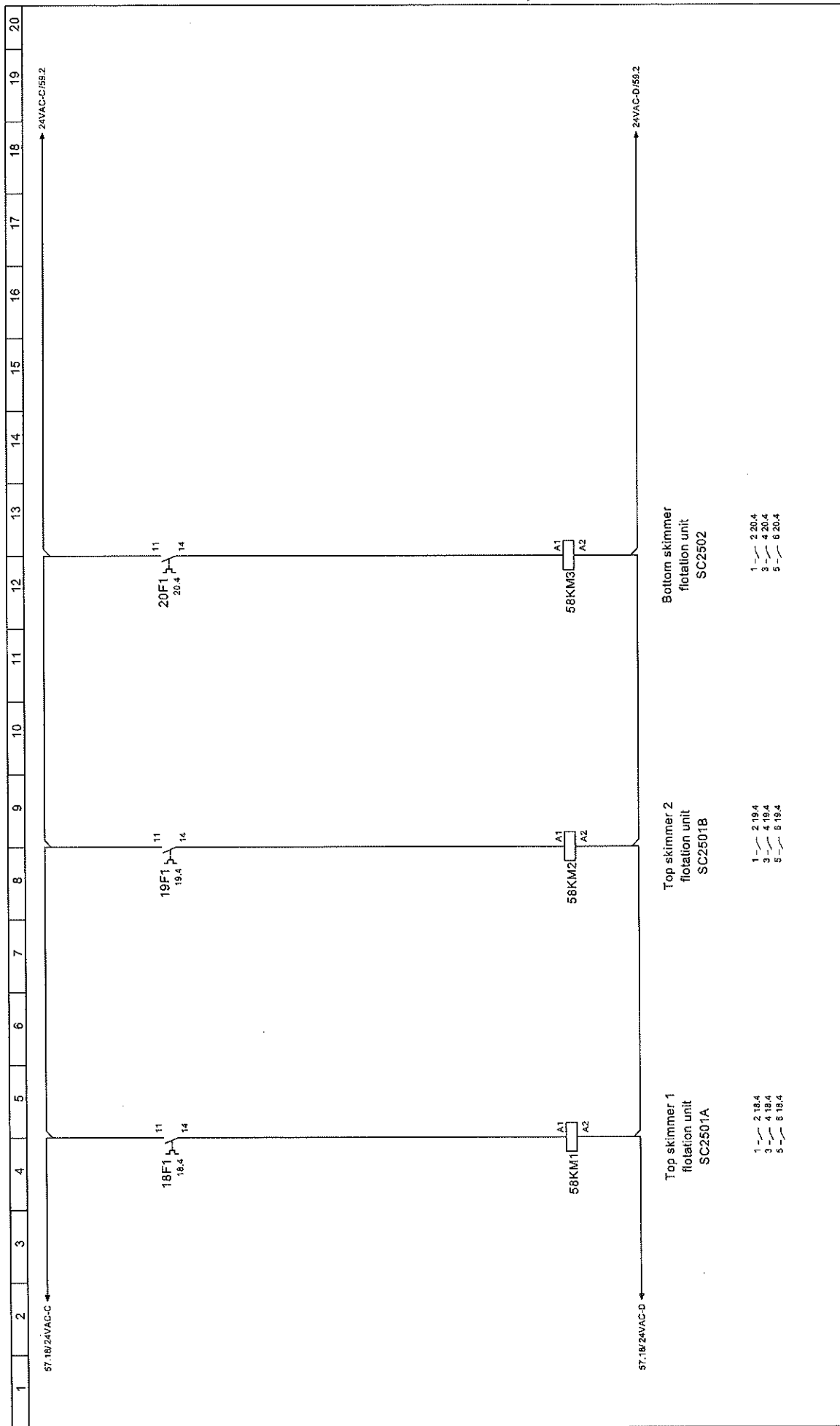
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Order : 112048.145	Remarks : PLC controlled panel	Revision by : PAV	Check : PAV
Engineer : KRM	Title : Control wiring	Revision date : 10.01.13	Page : 57 / 78

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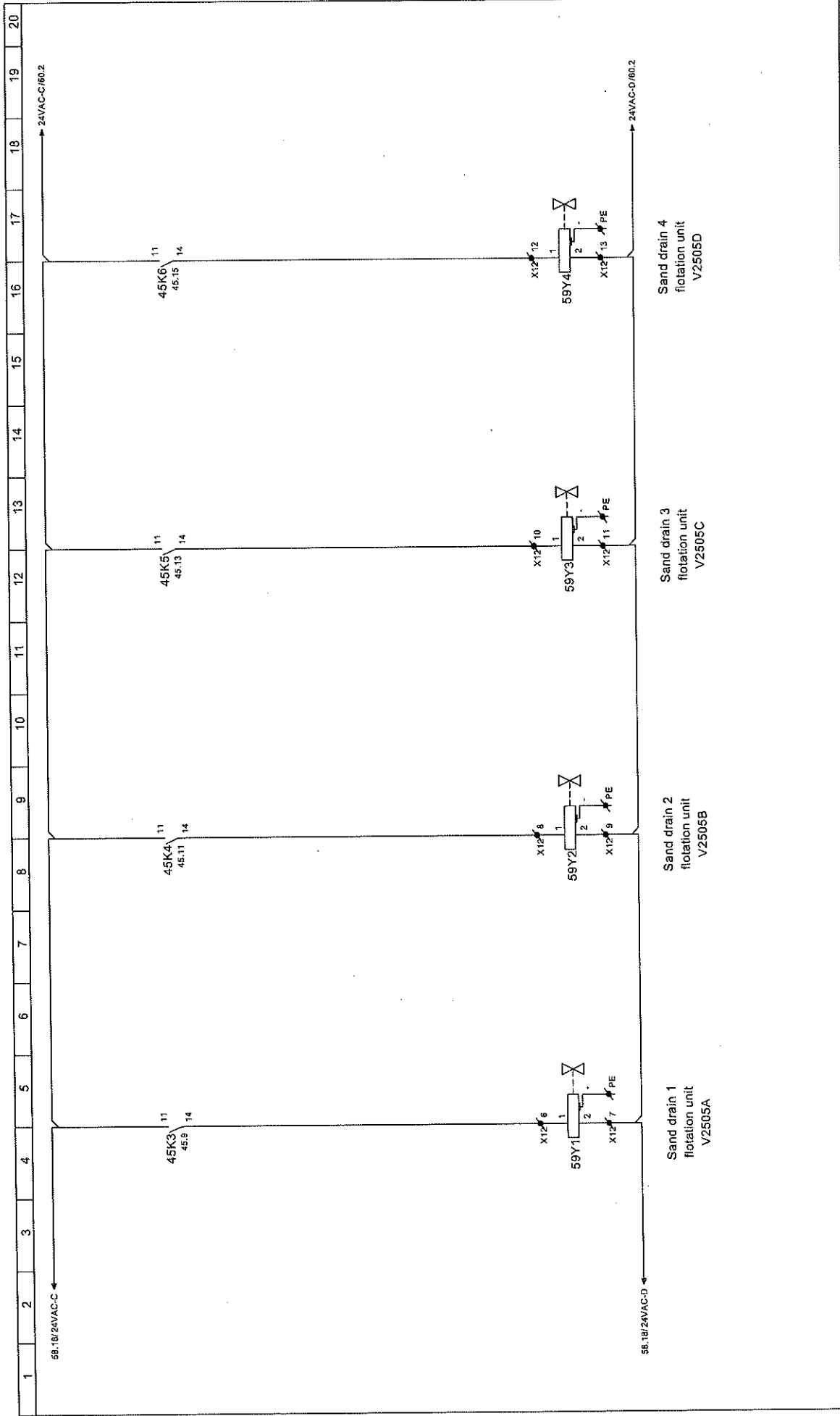
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Project : Austral Ilo - 112048		Revision : A	Created on : 17.07.06
Order : 112048.145	Remarks : PLC controlled panel	Revision by : PAV	Check : PAV
Engineer : KRM	Title : Control wiring	Revision date : 28.12.12	Page : 58 / 78



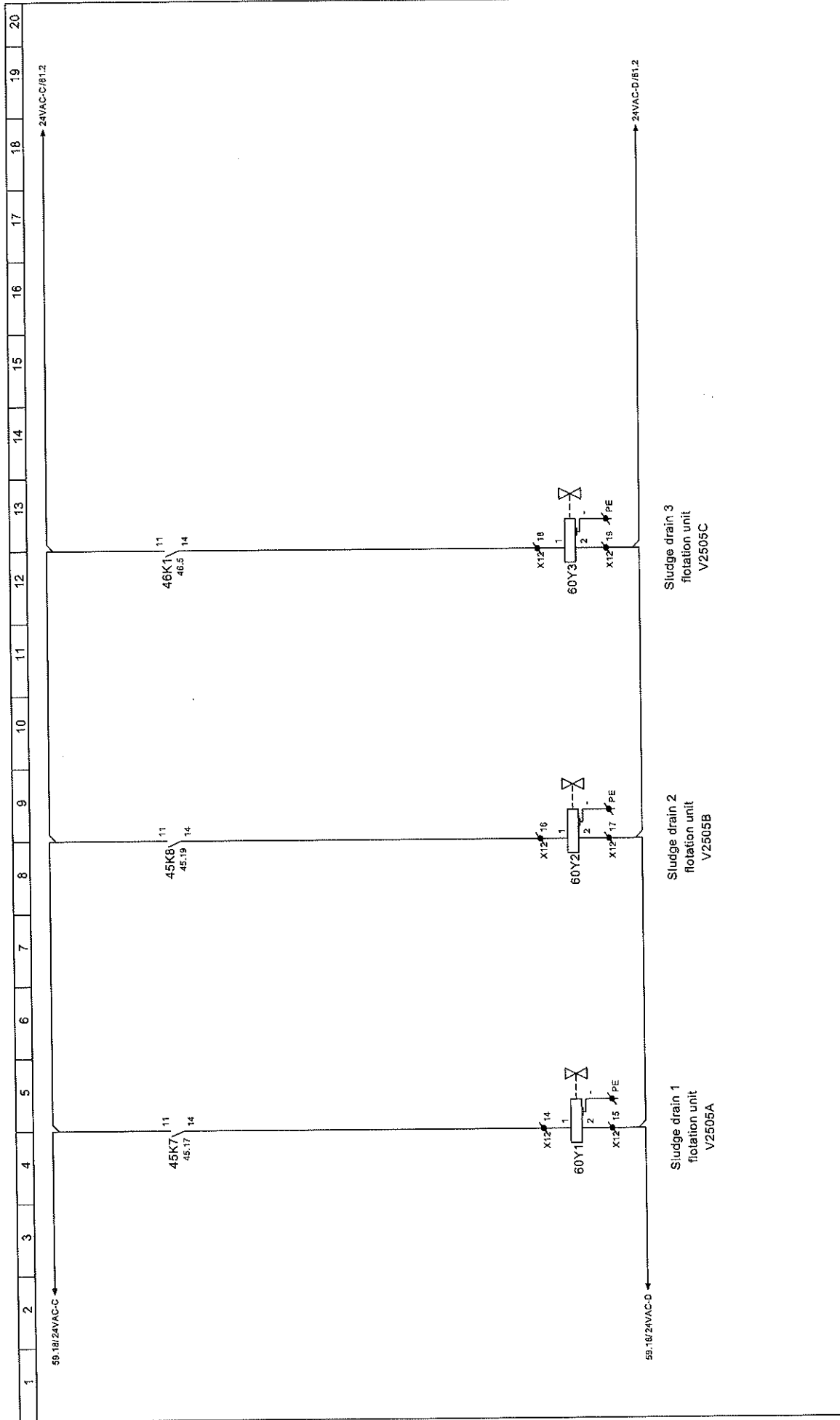
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Project : Austral Ilo - 112048		Revision : A	Created on : 17.07.06
Order : 112048.145	Remarks : PLC controlled panel	Revision by : PAV	Check : PAV
Engineer : KRM	Title : Control wiring	Revision date : 28.12.12	Page : 59 / 78



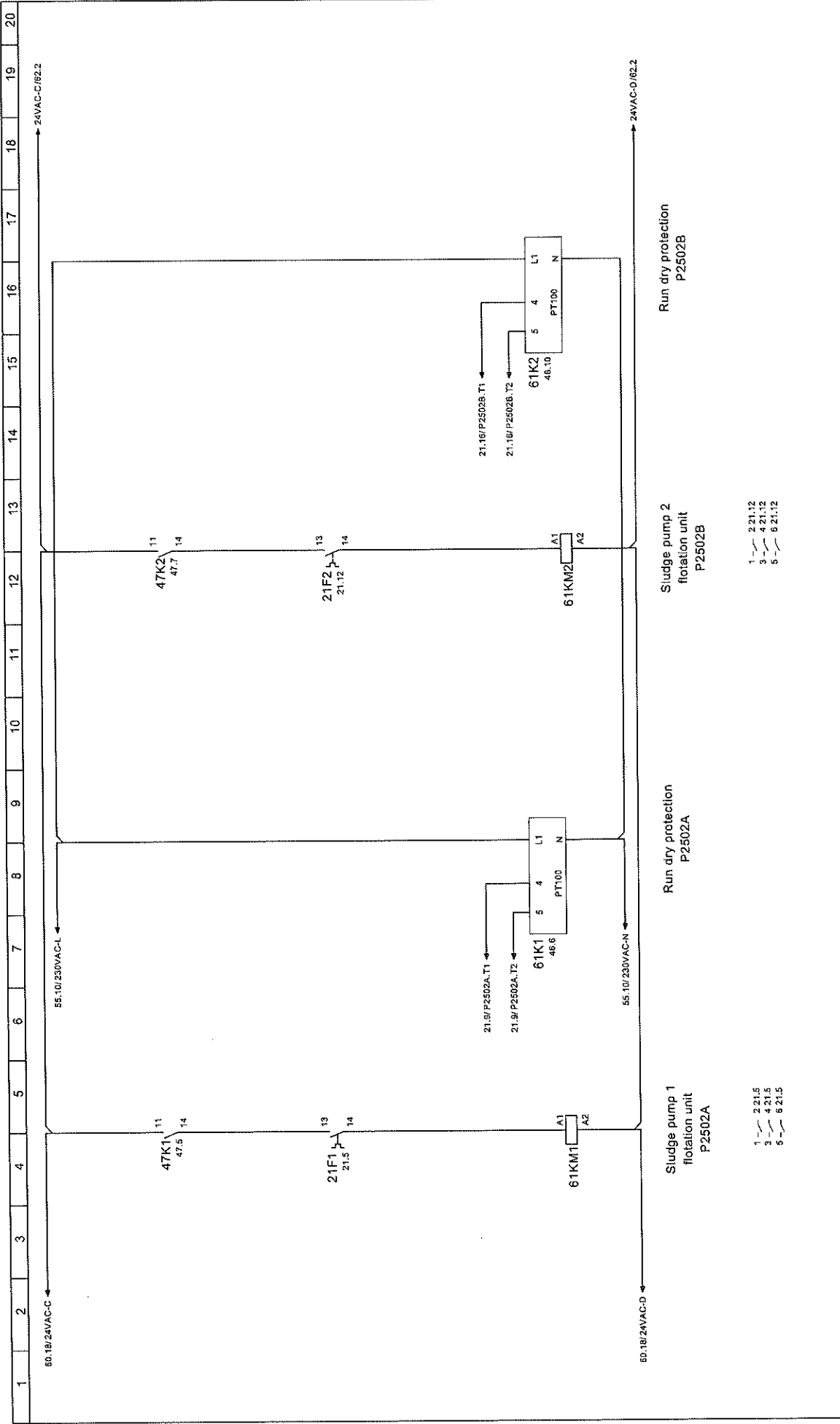
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Project : Austral Ilo - 112048		Revision : A	Created on : 17.07.06
Order : 112048.145	Remarks : PLC controlled panel	Revision by : PAV	Check : PAV
Engineer : KRM	Title : Control wiring	Revision date : 28.12.12	Page : 60 / 78



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Project : Austral Ilo - 112048		Revision : A	Created on : 17.07.06
Order : 112048.145	Remarks : PLC controlled panel	Revision by : PAV	Check : PAV
Engineer : KRM	Title : Control wiring	Revision date : 28.12.12	Page : 61 / 78



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1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
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61.18/24VAC-C ←

→ 24VAC-C/63.2

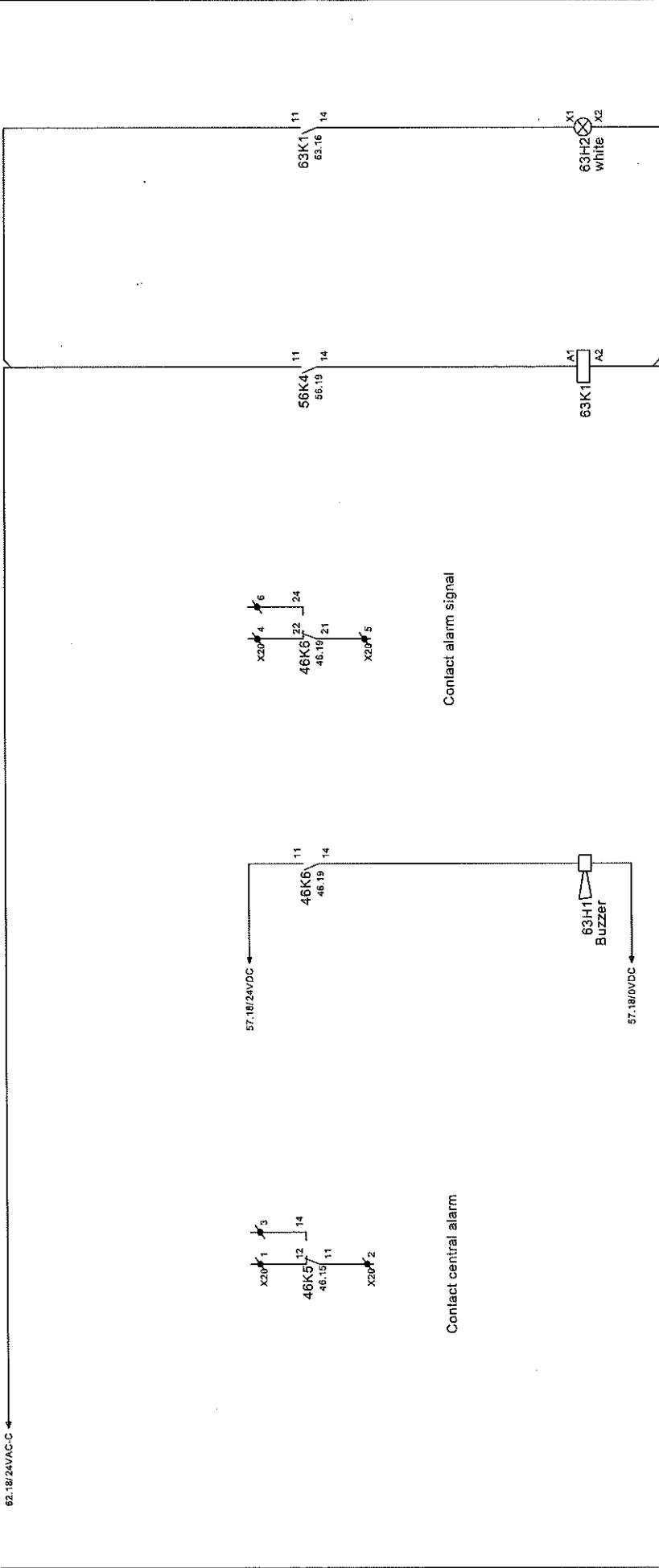
Spare page

61.18/24VAC-D ←

→ 24VAC-D/63.2

Project : Austral Ilo - 112048		Revision : A	Created on : 17.07.06	 VISION ON SUSTAINABILITY Copyright reserved www.nirhuuswater.com - T +31 (0)315 655 466
Order : 112048.145	Remarks : PLC controlled panel	Revision by : PAV	Check : PAV	
Engineer : KRM	Title : Control wiring	Revision date : 28.12.12	Page : 62 / 78	

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
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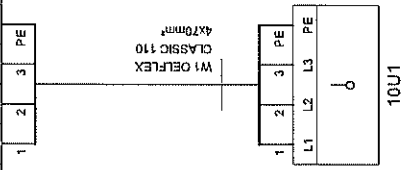
Control voltage
24VAC

11 — 14 63.19
21 — 24 48.14

Project : Austral Ilo - 112048		Revision : A	Created on : 17.07.06	nijhuis WATER TECHNOLOGY VISION ON SUSTAINABILITY Copyright reserved www.nijhuis-water.com · T +31 (0)315 655 466
Order : 112048.145	Remarks : PLC controlled panel	Revision by : PAV	Check : PAV	
Engineer : KRM	Title : Control wiring	Revision date : 28.12.12	Page : 63 / 78	

* Cable diameters are for indication only, NWT doesn't take any responsibility for it! See general instructions on page 6 for further details
 * Cables must be selected in conformance with power consumption, environmental conditions, method of installation, local regulations and conditions

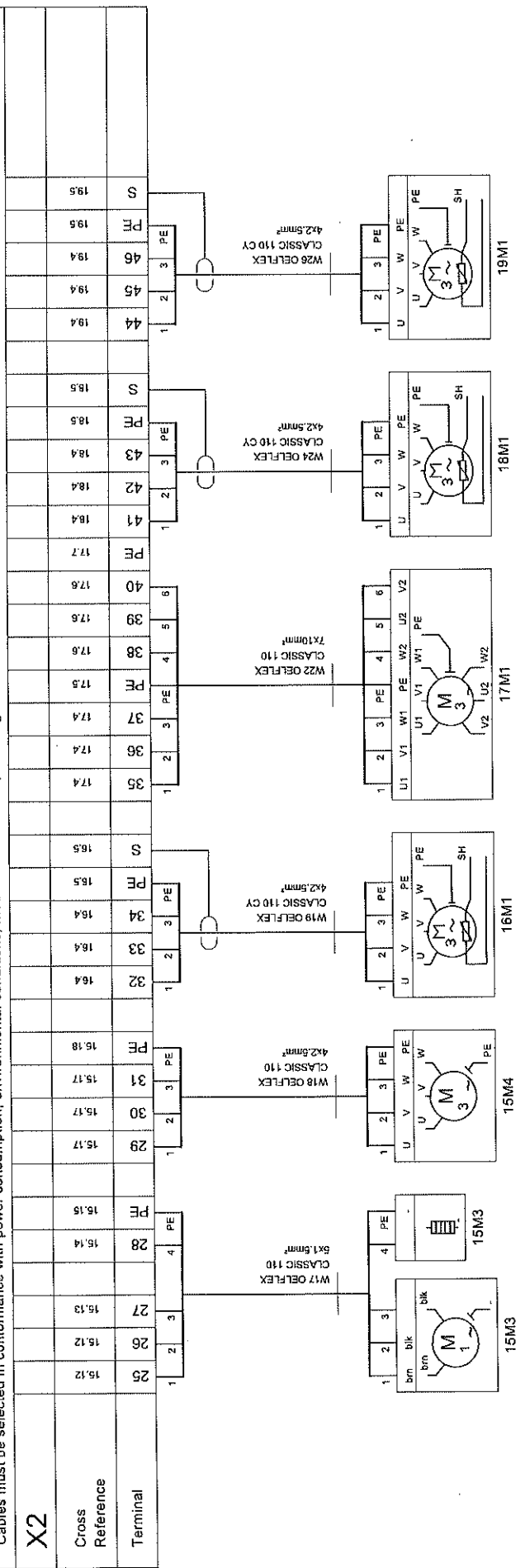
X1																				
Cross Reference	10.3	10.3	10.3	10.3	10.4															
Terminal	1	2	3	PE	PE															



L1 L2 L3 PE

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	
* Cable diameters are for indication only, NWT doesn't take any responsibility for it! See general instructions on page 6 for further details * Cables must be selected in conformance with power consumption, environmental conditions, method of installation, local regulations and conditions																				
X2																				
Cross Reference	11.5	11.5	11.6	11.7	11.7	11.8	11.8	11.9	11.10	12.5	12.6	13.5	13.6	14.5	14.6	15.4	15.4	15.8	15.9	
Terminal	1	2	3	PE	4	5	6	PE	7	8	PE	9	10	11	12	13	14	15	16	
	1	2	3	PE	4	5	6	1	2	PE	1	2	PE	1	2	PE	1	2	3	PE
	Feed pump flotation unit P0201 Influent control valve flotation unit V0203 Detection failure control valve V0203 Detection closed V0203 Dosing pump coagulant P1201 Dosing pump ferric P1401 Dosing pump neutraliser P1501 Mixer flocculant MX1801 Dosing screw flocculant DD1801																			
Project	Austral Ilo - 112048																			
Order	112048,145																			
Engineer	PAV																			
Remarks	PLC controlled panel																			
Title	Motor groups																			
Revision	A																			
Created on	17.07.06																			
Check	PAV																			
Page	65 / 78																			
Revision date	13.03.13																			
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* Cable diameters are for indication only, NWT doesn't take any responsibility for it! See general instructions on page 6 for further details
 * Cables must be selected in conformance with power consumption, environmental conditions, method of installation, local regulations and conditions

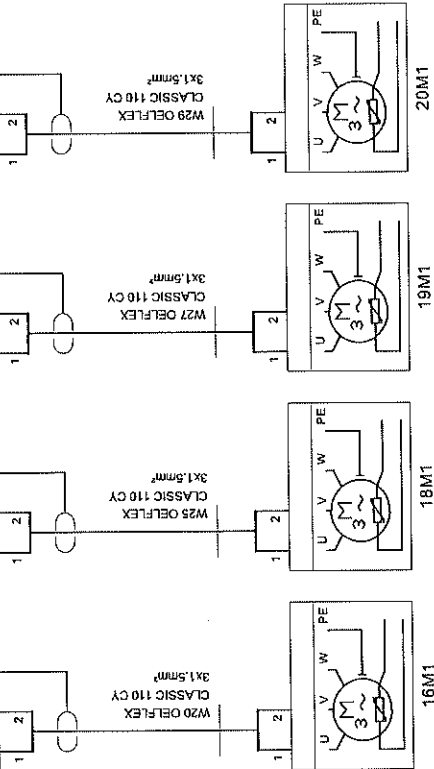


Project : Austral Ilo - 112048		Revision : A	Created on : 17.07.06		
Order : 112048.145	Remarks : PLC controlled panel	Revision by : PAV	Check : PAV	VISION ON SUSTAINABILITY Copyright reserved www.nijhuis-water.com - T +31 (0)315 655 466	
Engineer : PAV	Title : Motor groups	Revision date : 13.03.13	Page : 66 / 78		
Vibrator flocculant M1801		Top skimmer 1 flotation unit SC2501A			
Heater flocculant HE1801		Top skimmer 2 flotation unit SC2501B			
Transfer pump flocculant P1802					
Dosing pump flocculant P1801					
		Recirculation pump flotation unit P2501			

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20																																																															
* Cable diameters are for indication only, NWT doesn't take any responsibility for it! See general instructions on page 6 for further details * Cables must be selected in conformance with power consumption, environmental conditions, method of installation, local regulations and conditions																																																																																		
X4																																																																																		
<table><tr><td>Cross Reference</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td><td>14</td><td>15</td><td>16</td><td>17</td><td>18</td><td>19</td><td>20</td></tr><tr><td>Terminal</td><td>11.1.13</td><td>11.1.14</td><td>36.4</td><td>36.4</td><td>36.4</td><td>36.5</td><td>36.8</td><td>36.8</td><td>36.8</td><td>36.16</td><td>36.16</td><td>36.16</td><td>36.17</td><td>41.4</td><td>41.4</td><td>41.8</td><td>41.8</td><td>41.12</td><td>41.16</td><td>42.12</td></tr><tr><td></td><td>PE</td><td>PE</td><td>PE</td><td>PE</td><td>PE</td><td>PE</td><td>PE</td><td>PE</td><td>PE</td><td>PE</td><td>PE</td><td>PE</td><td>PE</td><td>PE</td><td>PE</td><td>PE</td><td>PE</td><td>PE</td><td>PE</td><td>PE</td></tr></table>																				Cross Reference	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Terminal	11.1.13	11.1.14	36.4	36.4	36.4	36.5	36.8	36.8	36.8	36.16	36.16	36.16	36.17	41.4	41.4	41.8	41.8	41.12	41.16	42.12		PE	PE	PE	PE	PE	PE	PE	PE	PE	PE	PE	PE	PE	PE	PE	PE	PE	PE	PE	PE
Cross Reference	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20																																																														
Terminal	11.1.13	11.1.14	36.4	36.4	36.4	36.5	36.8	36.8	36.8	36.16	36.16	36.16	36.17	41.4	41.4	41.8	41.8	41.12	41.16	42.12																																																														
	PE	PE	PE	PE	PE	PE	PE	PE	PE	PE	PE	PE	PE	PE	PE	PE	PE	PE	PE	PE																																																														
Position meas. valve V0203 Flow measuring influent water Flotation unit FIC0201 pH measuring flocculator QIC0801 Flow measuring sludge pumps Flotation unit FIC2502 Level measuring balance tank LC0201 Level measuring sludge compartment Level measuring sludge compartment Spare 2-wire [D] analog input Spare 4-wire [C] analog input																																																																																		
Project : Austral Ilo - 112048 Order : 112048.145 Engineer : PAV Remarks : PLC controlled panel Title : Analog input signals																																																																																		
Revision : A Revision by : PAV Revision date : 13.03.13 Created on : 17.07.06 Check : PAV Page : 69 / 78																																																																																		
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* Cable diameters are for indication only, NWT doesn't take any responsibility for it! See general instructions on page 6 for further details
 * Cables must be selected in conformance with power consumption, environmental conditions, method of installation, local regulations and conditions

X6																			
Cross Reference		16.7	16.7	16.8															
		1	2	5															
Terminal																			



PTC protection P1801
 PTC protection SC2501A
 PTC protection SC2501B
 PTC protection SC2502

Project : Austral Ilo - 112048

Order : 112048.145

Engineer : PAV

Revision : A

Created on : 17.07.06

Check : PAV

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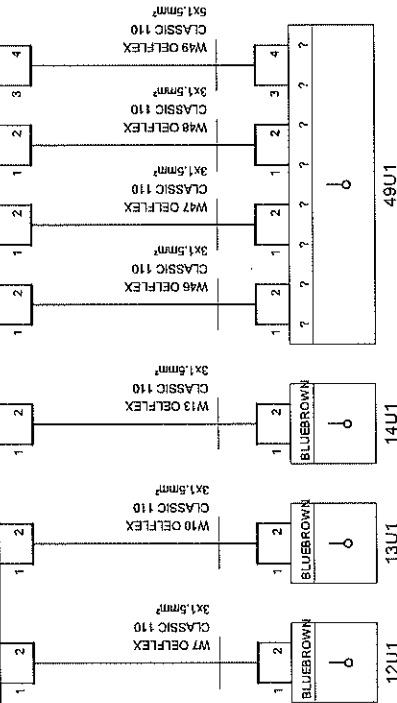
[illegible]

* Cable diameters are for indication only, NWT doesn't take any responsibility for it! See general instructions on page 6 for further details
 * Cables must be selected in conformance with power consumption, environmental conditions, method of installation, local regulations and conditions

X101

Cross
Reference

Terminal



Start/stop dosing pump P1201
 Start/stop dosing pump P1401
 Start/stop dosing pump P1501
 Spare
 Stop feeding balance tank due to high level LCO201 (To client)
 Enable start sludge pumps P2502A/B (From client)

Project : Austral Ilo - 112048

Order : 112048.145

Engineer : PAV

Remarks : PLC controlled panel

Title : External start/stop signals

Revision : A

Revision by : PAV

Revision date : 13.03.13

Created on : 17.07.06

Check : PAV

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